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10 **(a)** Planetary nebulae form as part of the life cycle of stars.

(i) Complete the sentence about a planetary nebula.

Stars expand when they run out of hydrogen to fuel nuclear reactions.

A ..... forms a planetary nebula. [1]

(ii) State what is at the centre of a planetary nebula.

..... [1]

(b) A planetary nebula emits blue light and red light at two specific wavelengths: 5.0×10^{-7} m and 6.6×10^{-7} m.

(i) Explain why the light with a wavelength of 5.0×10^{-7} m is the blue light.

.....

..... [1]

(ii) The speed of light in a vacuum is 3.0×10^{8} m/s.

Calculate the frequency of the blue light.

frequency = ... Hz [2]

[Total: 5]

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